

SLIDE 1



YOUR EC2 SERVER CRASHED AT 2 AM!



But...

USERS NOTICED NOTHING.

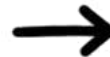
≡ HOW? ≡



USERS



APPLICATION



EC2 INSTANCE
(CRASHED)



INTRODUCING
AWS HIGH AVAILABILITY



Production systems should
EXPECT FAILURE – NOT FEAR IT.

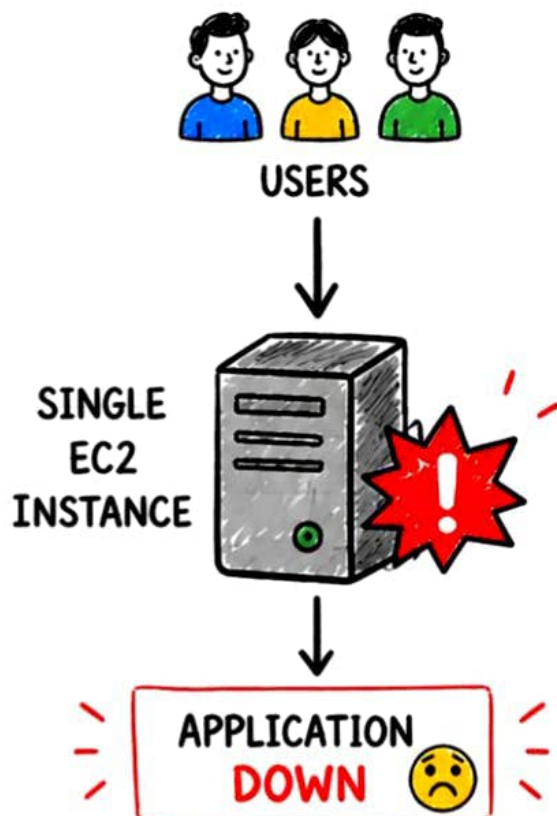


SLIDE 2

THE WRONG ARCHITECTURE



ONE SERVER = ONE BIG RISK



THE PROBLEMS

- ✗ Single EC2 instance
- ✗ Single point of failure
- ✗ Server crashes
→ application unavailable
- ✗ No automatic recovery
- ✗ Manual intervention required



Never depend on one EC2 instance for a critical production workload.

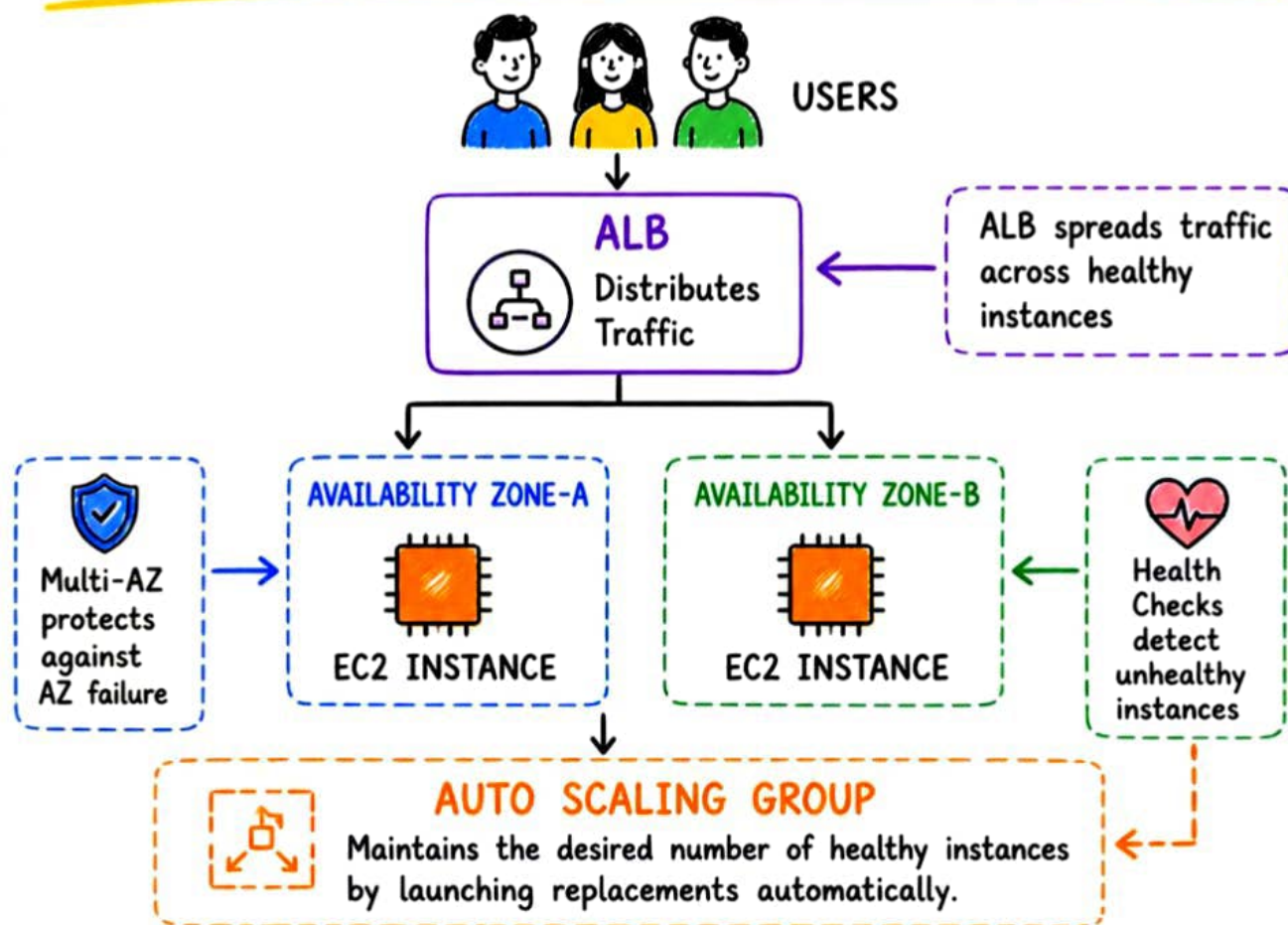


Availability is a design choice,
not a lucky outcome.

SLIDE 3



BUILD FOR FAILURE: THE HIGH-AVAILABILITY ARCHITECTURE



KEY COMPONENTS



ALB

Distributes traffic evenly



HEALTH CHECKS

Detect unhealthy instances



AUTO SCALING

Maintains required capacity



MULTI-AZ

Ensures high availability across zones



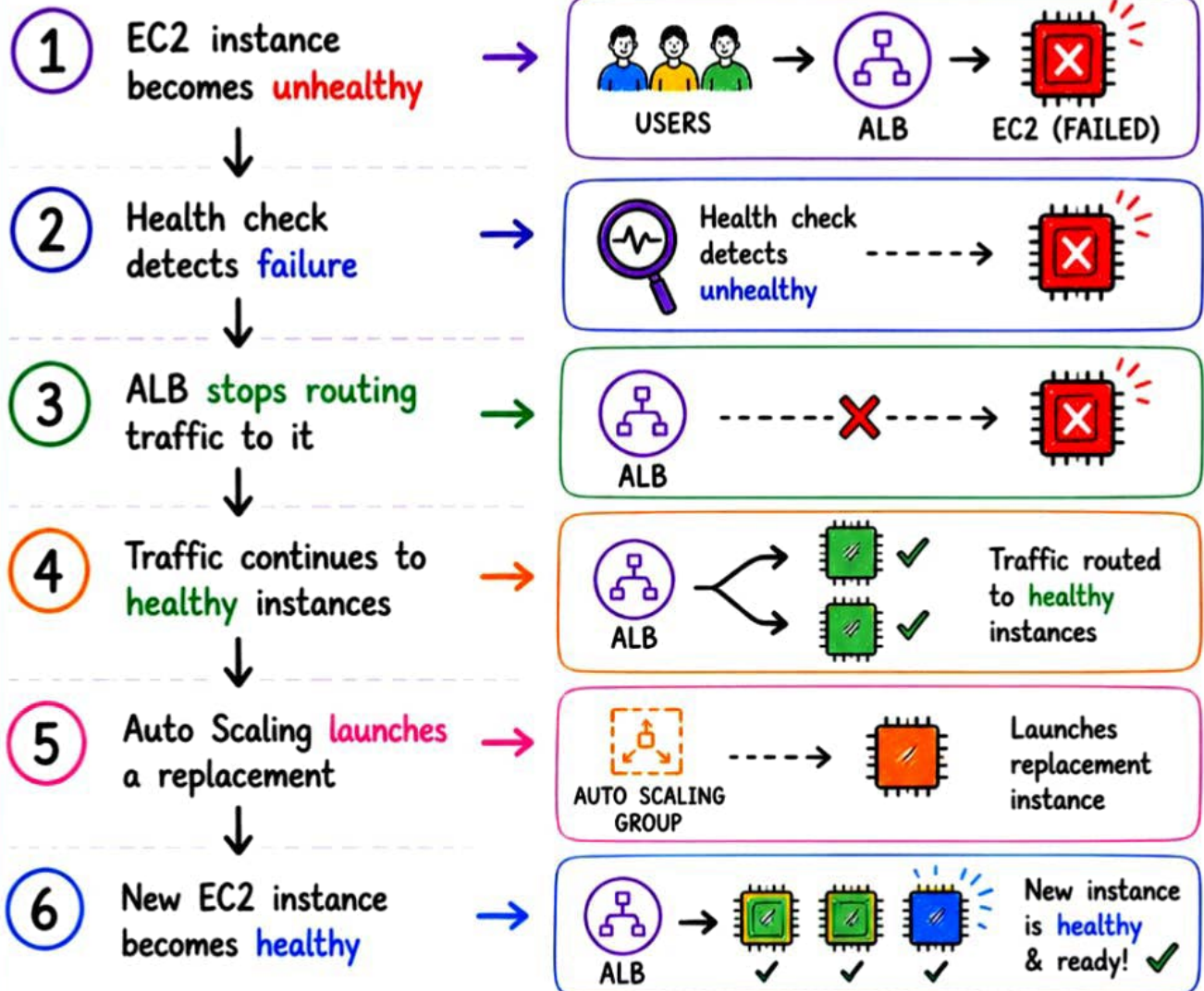
No single EC2 instance should own your availability.



SLIDE 4



EC2 FAILS. WHAT HAPPENS?



FAIL → **DETECT** → **REDIRECT** → **REPLACE** → **RECOVER**



High Availability is not about avoiding failure,
it's about recovering so fast that users never notice.



SLIDE 5



AUTO SCALING:

MIN vs DESIRED vs MAX

AUTO SCALING KEEPS YOUR APPLICATION
AVAILABLE & RESILIENT

MIN

Minimum capacity.
Never go below this.

2

DESIRED

Desired capacity.
Auto Scaling maintains this.

3

MAX

Maximum capacity.
Scale up to this limit.

6

HOW IT WORKS

NORMAL LOAD

Maintains desired capacity



= 3 INSTANCES

TRAFFIC SPIKE 🔥

Scales up automatically



= UP TO 6 INSTANCES

TRAFFIC DROPS

Scales down to desired



= BACK TO 3 INSTANCES

BENEFITS



High Availability
Always maintains
healthy capacity



Cost Optimized
Scale up/down based
on demand



Automatic Recovery
Replaces unhealthy
instances



Handles Traffic Spikes
Scale quickly during
high load



AUTO SCALING = SCALING + MAINTAINING
HEALTHY CAPACITY

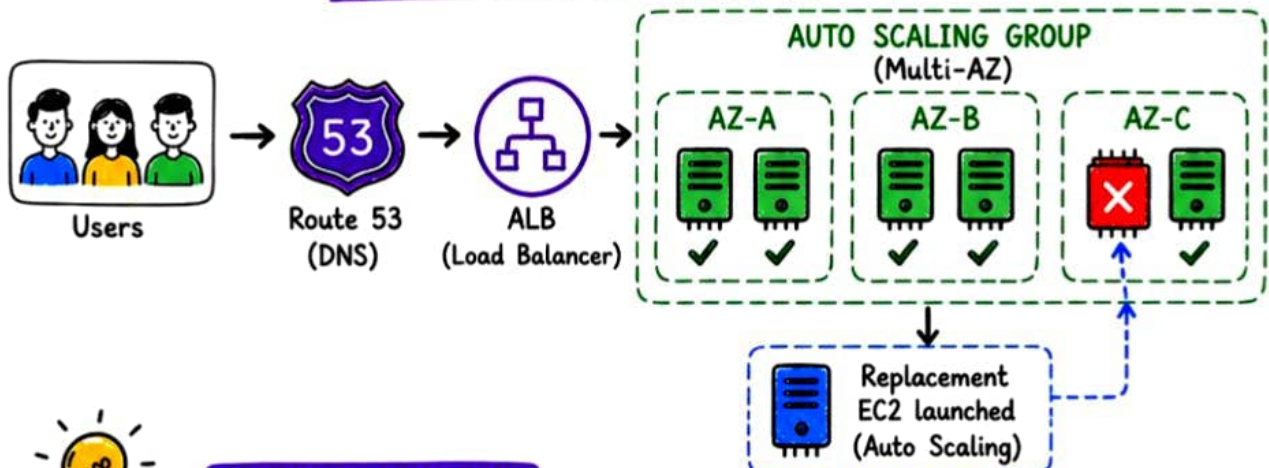


SLIDE 6



HOW DOES AWS SURVIVE AN EC2 FAILURE?

High Availability in Action!



INTERVIEW QUESTION:

One EC2 instance crashes in production.
How would you prevent an outage?



ANSWER:

Use an **Application Load Balancer (ALB)** with an **Auto Scaling Group (ASG)** spanning multiple Availability Zones.



- ✓ Health checks identify unhealthy instances.
- ✓ ALB automatically stops routing traffic to failed instances.
- ✓ Traffic continues to healthy instances.
- ✓ Auto Scaling launches new instances to maintain the desired capacity.



**DON'T DESIGN TO AVOID EVERY FAILURE.
DESIGN TO SURVIVE FAILURE.**

KEY TAKEAWAYS



ALB
Smart Traffic Routing



HEALTH CHECKS
Detect & Remove Unhealthy Instances



AUTO SCALING
Replaces & Maintains Capacity



MULTI-AZ
Built-in Resilience Across Zones



HIGH AVAILABILITY
Users Stay Happy, Always!



EXPECT FAILURE. PREPARE FOR FAILURE. KEEP USERS HAPPY.